

Phase-transition-driven ferroic response in 2D $\text{CuMnP}_2\text{Se}_6$ under ultra-low electric fields

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Low-field electric control of magnetic phase transitions is critical for the development of energy-efficient spintronic and non-volatile memory technologies. Yet, the weak magnetoelectric coupling in most known two-dimensional multiferroics hinders their practical implementation. Here, using crystal structure prediction and high-throughput first-principles calculations, we identify four previously unexplored bimetallic thio(seleno)phosphate multiferroics, $\text{XMnP}_2(\text{S/Se})_6$ ($X = \text{Cu}, \text{Au}$), all exhibiting robust in-plane spontaneous polarization—contrasting with the predominantly out-of-plane behavior in this material family—which effectively mitigates depolarization effects. In particular, $\text{CuMnP}_2\text{Se}_6$ hosts two stable C_2 -symmetric ferroelectric phases with opposite in-plane polarizations and distinct magnetic orders. Remarkably, an electric field as small as $\sim 0.001 \text{ V/\AA}$ can simultaneously reverse the polarization and induce an antiferromagnetic-to-ferromagnetic transition. The associated barrier is exceptionally low ($\sim 49 \text{ meV/f.u.}$), yielding a sizable magnetoelectric coefficient of $\sim 0.04 \text{ G} \cdot \text{cm/V}$. These results highlight a viable strategy for realizing electric-field-driven magnetism in intrinsic two-dimensional multiferroics under experimentally feasible conditions.

Electric-field control of magnetism offers a promising route to reduce energy consumption in spintronic and memory devices, paving the way for low-power, non-volatile information technologies that could transform future computing paradigms^{1–3}. To date, two primary mechanisms have been extensively investigated: (i) electric-field-induced magnetization switching^{4–6}, and (ii) electric-field-driven magnetic phase transitions^{7–10}. The former enables binary logic operations through spin-orbit torque-driven reversal of magnetization, whereas the latter involves switching between distinct magnetic phases—such as ferromagnetic (FM) and antiferromagnetic (AFM)—offering dynamic tunability between writable and interference-robust memory states. Realizing such functionality requires material systems where electric polarization and magnetic order are strongly coupled and reversibly tunable under experimentally accessible conditions.

Multiferroics, which simultaneously host two or more ferroic orders—such as magnetism, ferroelectricity, or ferroelasticity—within a single phase¹¹, offer a natural platform for realizing intrinsic magnetoelectric (ME) coupling. However, electric-field-induced magnetic phase transitions in existing multiferroics remain rare, primarily due to the generally weak coupling between electric and magnetic order parameters. As a result, polarization reversal alone is often insufficient to drive a magnetic transition^{12–22}. An alternative strategy leverages van der Waals heterostructures composed of 2D magnets and ferroelectrics, where electric control of magnetism arises from interfacial ME effects^{6,23–27}. However, such heterostructures often suffer from weak interfacial coupling and require precise lattice matching and stacking control, posing significant challenges for scalable fabrication and long-term device stability²⁸. Thus, identifying single-phase 2D multiferroics with strong intrinsic ME coupling has become a central

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pursuit in the field^{29–31}. Despite increasing interest, such materials remain scarce. Moreover, the substantial switching barriers (>several hundred meV/f.u.) observed in most reported materials demand high electric fields (>0.1 V/Å), creating a major bottleneck for their application in energy-efficient electronics.

In this study, we report four stable 2D bimetallic thio(seleno)phosphate (BMTP) multiferoics, $X\text{MnP}_2(\text{S/Se})_6$ ($X = \text{Cu}, \text{Au}$), each exhibiting two inequivalent C_2 -symmetric phases with opposite in-plane spontaneous polarizations, which helps mitigate depolarization fields. Notably, $\text{CuMnP}_2\text{Se}_6$ features two such C_2 phases with distinct magnetic ground states—FM and AFM. An ultralow electric field of $\sim 0.001 \text{ V/Å}$ can drive a structural transition between these phases, thereby reversing the polarization and switching the magnetic order simultaneously. The transition is characterized by a remarkably low switching barrier ($\sim 49 \text{ meV/f.u.}$) and a sizable ME coupling coefficient ($\sim 0.04 \text{ G} \cdot \text{cm/V}$). These findings demonstrate strong ME coupling and ultralow switching barriers in 2D BMTPs, highlighting their potential as a promising platform for electric-field control of magnetic phase transitions under experimentally accessible conditions.

Results

In-plane polarization in $\text{CuMnP}_2\text{Se}_6$

The BMTPs with the general formula $XYP_2(\text{S/Se})_6$ ($X, Y = \text{transition metals}$), crystallize in van der Waals layered structures and represent a versatile family of emerging piezoelectric, ferroelectric, and multiferoic materials^{32–35}. Their compositional flexibility—achieved by varying X and Y elements such as Cu, Ag, In, Cr, and V—enables systematic tuning of electronic, ferroelectric, and magnetic properties^{36–39}. Dimensional reduction to monolayers via mechanical exfoliation offers an effective pathway to tailor ferroic behaviors^{40–43}, with improved control over polarization and magnetic states.

A prototypical 2D BMTP structure, exemplified by CuCrP_2S_6 , consists of metal cations intercalated within a hexagonal framework composed of ethane-like $[\text{P}_2\text{S}_6]^{4-}$ units with C_3 symmetry (Fig. 1a). The Cu^{1+} and Cr^{3+} cations alternately occupy interstitial sites, forming a

secondary honeycomb sublattice. Each Cr^{3+} ion is centered within an octahedron of six S^{2-} ligands, accounting for the magnetic behavior, while the out-of-plane displacement of Cu^{1+} ion gives rise to spontaneous polarization⁴⁴. The out-of-plane (OOP) polarization switching proceeds through a nonpolar (NP) intermediate phase which possesses D_3 symmetry (Fig. 1b). Notably, a broad range of 2D BMTPs can be derived by substituting different transition-metal species into this structural motif^{36–39,45}.

To further explore the structural landscape of 2D BMTP family, we performed ab initio structure searches for $\text{CuMnP}_2\text{Se}_6$ using the CALYPSO method^{46–51}. We selected Mn as the Y -site transition metal because Mn^{3+} possesses a high-spin d^4 electron configuration. In an MnSe_6 octahedral environment, the Mn^{3+} ion adopts a $t_{2g}^3 e_g^1$ configuration, which is well known to produce a strong Jahn-Teller instability (Supplementary Fig. 1). The singly occupied e_g orbital (d_{z^2} or $d_{x^2-y^2}$) directly overlaps with the ligand p orbitals, leading to pronounced structural distortion of MnSe_6 octahedra, involving tetragonal compression or tetragonal elongation (Supplementary Table 1). This situation stands in sharp contrast to other 2D BMTPs such as $\text{CuCrP}_2\text{Se}_6$ (Cr^{3+} , t_{2g}^3), CuVP_2Se_6 (V^{3+} , t_{2g}^2) and $\text{CuFeP}_2\text{Se}_6$ (Fe^{3+} , $t_{2g}^3 e_g^2$), where the transition-metal ions are Jahn-Teller inactive. Consequently, the strong Jahn-Teller activity of Mn^{3+} may facilitate the emergence of new structural prototypes that are inaccessible to other transition metals.

As expected, two low-energy polar prototypes with C_2 symmetry were identified (Fig. 1c, d), referred to as the D-type and C-type phases, characterized by the formation of Cu-Mn dimer-like units and -Cu-Mn-Cu-Mn- chain-like configurations, respectively. In contrast to conventional BMTP structures, where the X cation typically displaces vertically from the center of its octahedron (Fig. 1a), the X ion in both newly identified phases remains centered, accompanied by slight in-plane displacements (insets of Fig. 1c, d). These displacements give rise to spontaneous in-plane polarization. Specifically, the D-type and C-type structures exhibit opposite in-plane polarization directions along $[\bar{1}10]$ and $[110]$, respectively. This behavior stands in sharp contrast to most

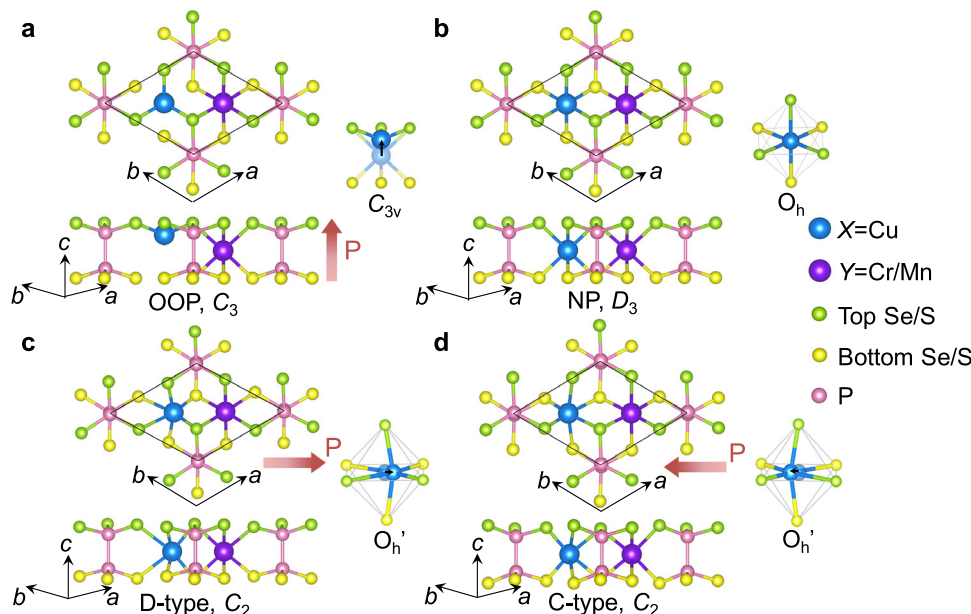


Fig. 1 | Crystal structures of four structural prototypes of 2D $XYP_2(\text{S/Se})_6$ viewed from the top and side. a Out-of-plane (OOP) ferroelectric phase with C_3 symmetry. **b** Nonpolar (NP) phase with D_3 symmetry. **c** In-plane ferroelectric phase with C_2 symmetry, polarized along the $[\bar{1}10]$ direction (D-type). **d** In-plane ferroelectric phase with C_2 symmetry, polarized along the $[110]$ direction (C-type). Blue

and purple spheres represent X and Y metal atoms, respectively; green and yellow spheres correspond to S/Se atoms on the top and bottom layers, respectively; pink spheres denote P atoms. Red arrows indicate the polarization directions. Four insets highlight the distinct coordination environments, local symmetries, and off-center displacement of the Cu ions.

previously reported 2D BMTP multiferroics, which typically display OOP polarization^{36,44}.

The distinct polarization behavior in these systems arises from two key factors: Jahn-Teller activity and symmetry-forbidden s - d coupling⁴⁴. The Jahn-Teller active Mn^{3+} ion ($t_{2g}^3 e_g^1$) lifts the e_g orbital degeneracy, leading to tetragonal compression in the D-type phase and tetragonal elongation in the C-type phase (Supplementary Table 1). These distortions generate an in-plane anisotropic local potential, which drives lateral Cu displacement. In contrast, in BMTP compounds such as CuCrP_2S_6 , where the Jahn-Teller effect is weak, the nearly symmetric crystal field fails to create such a gradient potential, thereby suppressing in-plane polarization.

Meanwhile, the local approximate O_h symmetry around the Cu^+ ion in $\text{CuMnP}_2\text{Se}_6$ forbids s - d coupling because the s and d orbitals belong to different irreducible representations (Supplementary Table 2). This symmetry-forbidden s - d hybridization contrasts with the symmetry-allowed s - d coupling in CuInP_2S_6 , which drives the OOP polarization⁴⁴. This forbidden s - d coupling in $\text{CuMnP}_2\text{Se}_6$ is further supported by the projected density of states (PDOS) and band-decomposed charge density (Supplementary Fig. 2). Therefore, the combination of strong Jahn-Teller distortion and symmetry-forbidden s - d coupling collectively determines the emergence and stabilization of in-plane polarization in $\text{CuMnP}_2\text{Se}_6$. A more detailed discussion is provided in the Supplementary Information.

High-throughput screening and identification of in-plane ferroelectric phases and magnetic ground states

Having identified four structural prototypes, we employed high-throughput first-principles calculations to systematically investigate the possibility of D-type and C-type structures within this material family. As illustrated in Fig. 2a, we considered compounds with the formula $\text{XYP}_2(\text{S/Se})_6$ ($X = \text{Cu, Ag, Au}$; $Y = \text{Mn, Tc}$), generated via targeted elemental substitution. The X-site elements (Cu, Ag, Au) were chosen as monovalent cations owing to their good displacive nature and can easily displace from the octahedral center, thus generating ferroelectric polarization. The Y-site transition metals were selected to ensure a high-spin $t_{2g}^3 e_g^1$ configuration (e.g., Mn^{3+} and Tc^{3+}), which naturally leads to Jahn-Teller activity and strong coupling among orbital, lattice, and spin degrees of freedom.

A total of 48 candidate structures were initially generated based on the four structural prototypes shown in Fig. 1. After full structural relaxation, 5 OOP-polarized phases (Fig. 1a) were found to be unstable and spontaneously transformed into NP phases. The potential energy profile of Cu-ion displacement further confirms the energetic preference for the NP phase (Supplementary Fig. 3). Consequently, the final dataset comprises 43 stable configurations: 7 OOP, 12 NP, 12 D-type, and 12 C-type phases (Fig. 2a). Among these, all $\text{XYP}_2(\text{S/Se})_6$ compounds in the D-type and C-type phases, except $\text{AgMnP}_2\text{Se}_6$, possess lower energies than their NP and OOP counterparts (Supplementary Table 3). Because the total energies of $\text{AgMnP}_2\text{Se}_6$ in different prototypes are nearly degenerate, indicating a structurally marginal system, it was excluded from subsequent phonon calculations.

Phonon spectrum calculations reveal that $\text{CuMnP}_2(\text{S/Se})_6$ and $\text{AuMnP}_2(\text{S/Se})_6$ are dynamically stable in both D-type and C-type configurations, whereas AgMnP_2S_6 is stable only in the C-type configuration (Supplementary Figs. 4, 5). We then determined the magnetic ground states of the dynamically stable compounds, excluding AgMnP_2S_6 since only its C-type phase is stable and no meaningful comparison between two polar phases can be made. For the remaining systems, we considered one FM and two possible AFM configurations (Supplementary Fig. 6). Interestingly, unlike other compounds where both polar phases prefer AFM ordering, $\text{CuMnP}_2\text{Se}_6$ displays two polar phases with different magnetic ground states—one FM and one AFM (Fig. 2b). This unique behavior allows an electric-field-induced polarization reversal to simultaneously trigger a magnetic phase transition between FM and AFM states. Consequently, we focus our subsequent discussion on $\text{CuMnP}_2\text{Se}_6$.

Distinct structural motifs, electronic properties and magnetic ground states in D-type and C-type $\text{CuMnP}_2\text{Se}_6$

Figure 3a and b show schematic structural diagrams of the D-type and C-type phases of $\text{CuMnP}_2\text{Se}_6$ in a $2 \times 2 \times 1$ supercell, with the P and Se atoms omitted for clarity. In the D-type phase, Cu and Mn atoms form dimer-like units with a nearest-neighbor distance of 3.43 Å, and the distance between neighboring dimers is 3.82 Å. In the C-type phase, Cu and Mn atoms form chain-like units with a nearest-neighbor distance

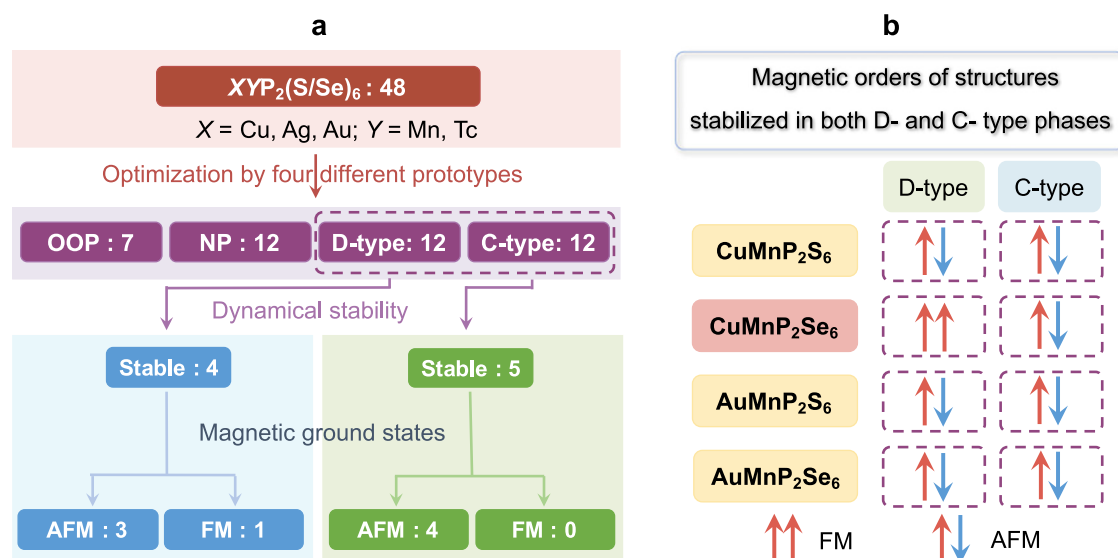


Fig. 2 | High-throughput computational screening. **a** Schematic workflow for the discovery of dynamically stable ferroelectric-ferromagnetic $\text{XYP}_2(\text{S/Se})_6$ compounds, where $X = \text{Cu, Ag, or Au}$ and $Y = \text{Mn or Tc}$, based on the four structural configurations shown in Fig. 1. “OOP” and “NP” denote the out-of-plane ferroelectric and nonpolar

phases, respectively. **b** Magnetic ground states of $\text{XYP}_2(\text{S/Se})_6$ ($X = \text{Cu, Au}$) summarized in a tabular format. Orange and blue arrows represent spin-up and spin-down magnetic moments, respectively.

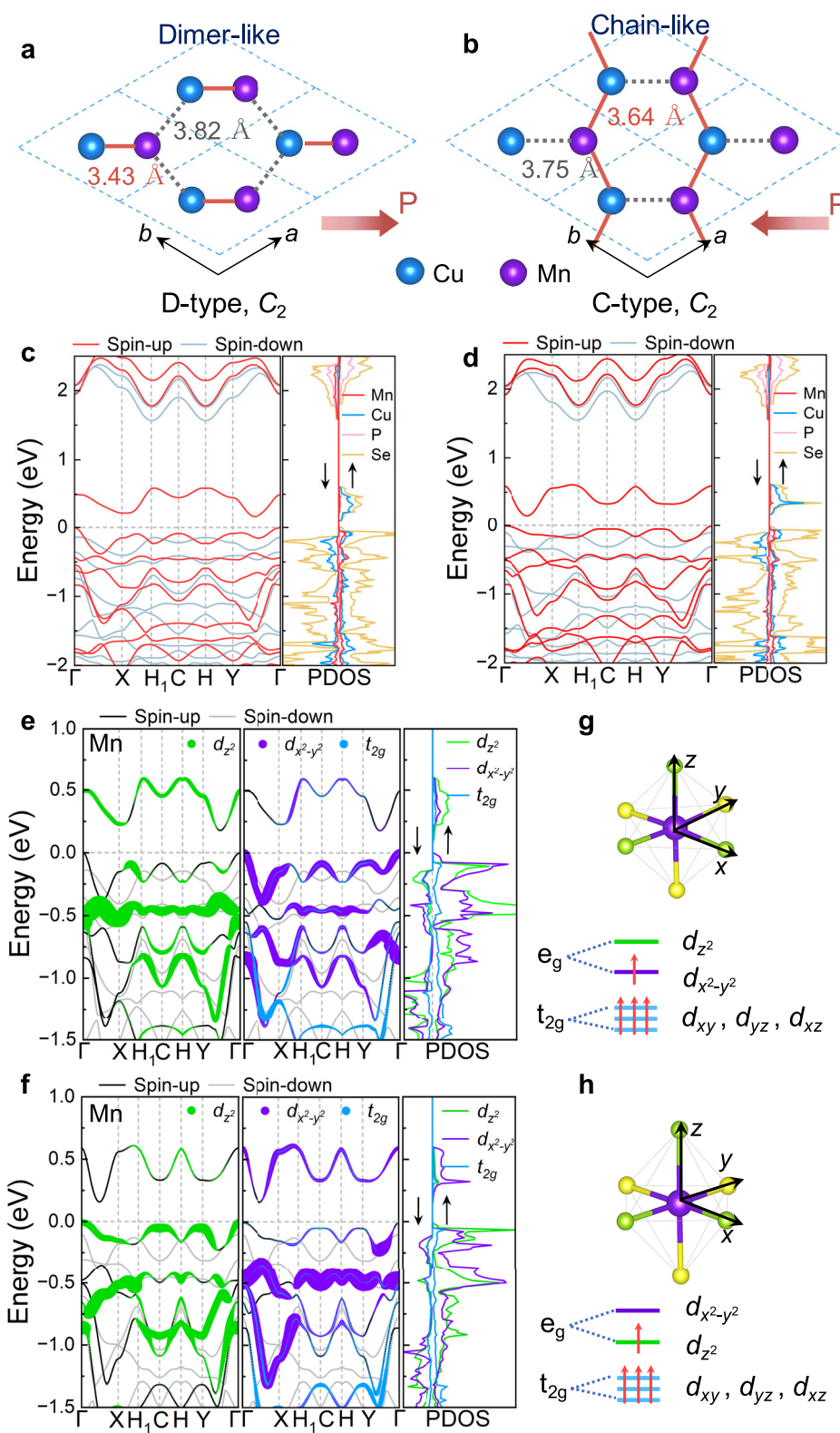


Fig. 3 | Schematic structural diagrams and electronic properties of 2D $\text{CuMnP}_2\text{Se}_6$. **a, b** Top view of the D-type and C-type phases in a 2×2 supercell. P and Se atoms are omitted for clarity to highlight the Cu-Mn dimer-like units and the -Cu-Mn-Cu-Mn- chain-like units. Blue and purple spheres denote Cu and Mn atoms,

respectively. **c, d** Spin-polarized band structures and projected density of states (PDOS) for the D-type and C-type phases. **e, f** Orbital-resolved spin-up band structure projected onto Mn d orbitals for the D-type and C-type phases. **g, h** Mn d^4 orbital energy levels with electron occupation for the D-type and C-type phases.

of 3.64 Å along the chain, and the distance between neighboring chains is 3.75 Å. The C-type phase is slightly lower in energy than the D-type phase by ~4 meV/f.u.

The cohesive energy of CuMnP₂Se₆ is calculated using the equation $E_{\text{coh}} = (E_{\text{tot}} - E_{\text{Cu}} - E_{\text{Mn}} - 2E_{\text{P}} - 6E_{\text{Se}})/10$, where E_{tot} is the total energy of 2D CuMnP₂Se₆, and E_{Cu} , E_{Mn} , E_{P} , and E_{Se} are the total energies of isolated Cu, Mn, P, and Se atoms, respectively. The cohesive energies for both D-type and C-type phases are similar, ~-3.92 eV/atom, suggesting that both structures are thermodynamically stable and potentially synthesizable. Furthermore, Ab initio molecular dynamics simulations at 100 K confirm the thermal stability and in-plane long-range polarized order of CuMnP₂Se₆ (Supplementary Fig. 7).

The spin-polarized band structures show that both the D-type and C-type phases are semiconductors with indirect band gaps of 0.18 eV and 0.16 eV, respectively (Fig. 3c, d). Spin-orbit coupling has a negligible effect on the electronic structure and is therefore excluded from subsequent calculations (Supplementary Fig. 8). Spin density calculations show that the spin moments are mainly localized on Mn ions (Supplementary Fig. 9). The PDOS confirms that the spin polarization near the Fermi level mainly comes from Mn 3d orbitals. The formal spin moments for Mn and Cu are 4 μ_B and 0 μ_B , respectively, consistent with their expected valence states of Mn³⁺ and Cu¹⁺. The overlap between the Mn 3d and Cu 3d orbitals is observed below the Fermi level in PDOS (Supplementary Fig. 10). In fact, the overlap originates from the strong hybridization of Mn 3d-Se 4p and Cu 3d-Se 4p states, making the Cu and Mn d states appear to overlap.

In the D-type phase, the valence band maximum is primarily derived from Mn $d_{x^2-y^2}$ orbitals, while the conduction band minimum mainly originates from Mn d_{z^2} orbitals (Fig. 3e). In contrast, the C-type phase shows the opposite orbital contributions to the valence and conduction bands (Fig. 3f). The calculated onsite energies of Mn d orbitals (Supplementary Table 4) confirm that the $d_{x^2-y^2}$ orbital lies lower in energy than the d_{z^2} orbital in the D-type phase, whereas the reverse holds for the C-type phase. This distinct orbital ordering enhances Mn $d_{x^2-y^2}$ -Se p_{xy} hybridization in the D-type phase and Mn d_{z^2} -Se p_z hybridization in the C-type phase (Fig. 3g, h; Supplementary Fig. 11). Structurally, the D-type phase exhibits tetragonal compression, which energetically favors the formation of Mn-Cu dimer-like units, whereas the C-type phase shows tetragonal elongation, stabilizing a chain-like -Mn-Cu-Mn-Cu- arrangement (Supplementary Table 1; Fig. 3a, b). This structural response reflects the system's tendency to minimize total energy under different orbital hybridization conditions. In turn, these distinct structural configurations tend to modulate the Mn-Mn exchange pathways, which may lead to different magnetic ground states in the two phases.

The spin Hamiltonian is defined as $H = -\sum_{i,j} J_1 \vec{S}_i \cdot \vec{S}_j - \sum_{i,k} J_2 \vec{S}_i \cdot \vec{S}_k$, where J_1 and J_2 are the nearest-neighbor and next-nearest neighbor magnetic exchange parameters between Mn ions, respectively^{22,52–54}. The summation i, j , and k run over all the Mn sites. S is the formal magnetic moment of Mn ion. The positive and negative J values represent the FM and AFM coupling, respectively. By energy mapping analysis on the considered magnetic configurations (Supplementary Fig. 6), the calculated J_1 and J_2 for the D-type are 4.70 meV and 1.30 meV, respectively, indicating the FM ground state. In contrast, the calculated J_1 and J_2 for the C-type are 4.67 meV and -4.72 meV, respectively, indicating the AFM ground state. Monte Carlo simulations show that the magnetic critical temperature for FM D-type and AFM C-type phases are ~31 K and ~18 K, respectively (Supplementary Fig. 12).

We further investigated the impact of the Hubbard U parameter on magnetic ordering and orbital-level splitting for both D-type and C-type phases, finding that the overall qualitative trends and key conclusions are robust over a broad range of U values (Supplementary Tables 5 and 6).

Electric-field-induced magnetic switching in CuMnP₂Se₆

Berry phase calculations yield an in-plane polarization of 0.9×10^{-10} C/m along the $[\bar{1}10]$ direction for the D-type phase and 0.7×10^{-10} C/m along the $[\bar{1}10]$ direction for the C-type phase (Supplementary Figs. 13, 14). Born effective charge analysis^{55,56} gives corresponding values of 0.4×10^{-10} C/m for the D-type phase and 0.2×10^{-10} C/m for the C-type phase, in good agreement with those obtained from the Berry-phase method. These values are larger than those reported for other DTMPs^{12,36,57,58}, such as AgBiP₂Se₆ (1.2×10^{-12} C/m), CuCrP₂S₆ (0.79×10^{-12} C/m), AuZrP₂S₆ (9.74×10^{-12} C/m), and comparable to other 2D multiferroics^{13,16,59–61}, such as (CrBr₃)₂Li (0.92×10^{-10} C/m), and γ -ALOOH (0.85×10^{-10} C/m).

The polarization in both the D-type and C-type phases can only be switched by 120° through an in-plane rotation of 120° or 240° (Fig. 4a, b; Supplementary Fig. 15a, b). This process can also be understood as a lateral displacement of Cu ions. The corresponding energy barriers for ferroelectric switching are ~36 meV/f.u. for the D-type phase and ~32 meV/f.u. for the C-type phase. In contrast, a 180° polarization reversal within the same phase (either D-type or C-type) is prohibited, as the positions of Cu and Mn ions cannot be interchanged directly.

However, 180° polarization reversal remains feasible through an electric-field-induced phase transition between the D-type and C-type structures, accompanied by the migration or removal of Cu atoms (Fig. 4c; Supplementary Fig. 15c). Specifically, reversing the applied electric field—e.g., from $[\bar{1}10]$ to $[\bar{1}\bar{1}0]$ —can trigger a transition between the two phases, owing to their opposite polarization orientations. This electric-field-driven structural transformation simultaneously induces a magnetic phase transition, owing to the distinct magnetic ground states of the D-type and C-type structures. Figure 4c illustrates the energy barrier associated with this phase transition, which proceeds via an intermediate structure that closely resembles the D_3 configuration and can be regarded as the NP phase (Supplementary Fig. 16). The calculated barriers for both D-to-C and C-to-D transitions are nearly identical, ~49 meV/f.u. This value is nearly an order of magnitude lower than those reported for other 2D multiferroics exhibiting electric-field-induced magnetic transitions^{29–31}. Notably, the D-to-C transition can also be triggered by applying tensile strain (Supplementary Fig. 17).

The ME coupling coefficient can be estimated as $\alpha \approx \mu_0 \Delta M / E_c$, where μ_0 is the vacuum permeability, ΔM is the magnetization change per unit cell associated with polarization switching, and E_c is the coercive electric field⁶². As is well known, a direct evaluation of E_c from first-principles is very challenging, since in real ferroelectric materials, polarization reversal is typically dominated by domain-wall motion and defect-mediated processes rather than homogeneous switching, and E_c is also strongly temperature dependent⁶³. Therefore, we adopted a more reasonable reference value based on experimentally accessible ferroelectric systems to estimate E_c . The values of energy barrier (4.9 meV/atom) and polarization (13 $\mu\text{C}/\text{cm}^2$ for the D-type phase and 9 $\mu\text{C}/\text{cm}^2$ for the C-type phase, assuming an effective thickness of 7 Å) in CuMnP₂Se₆ are comparable to those reported for ferroelectric Hf(Zr)_{1-x}O₂ thin films (7.3 meV/atom; 22.5 $\mu\text{C}/\text{cm}^2$)⁶⁴. Thus, we adopt the experimentally reported coercive field of Hf(Zr)_{1-x}O₂, 65 kV/cm, to estimate the ME coupling coefficient of CuMnP₂Se₆. This yields $\alpha \approx 0.04$ G · cm/V, which exceeds those of bulk multiferroic Fe/BaTiO₃ (~0.01 G · cm/V)⁶², and 2D WReCl₆ (~0.02 G · cm/V)²⁹, indicating the strong ME coupling.

To evaluate the feasibility of in-plane electric-field-driven structural transitions between the D-type and C-type phases, we investigate the field-dependent total energies of both configurations (Fig. 4d). The two phases exhibit opposite responses: an electric field along the $[\bar{1}10]$ direction stabilizes the C-type phase, while a field along $[\bar{1}\bar{1}0]$ favors the D-type phase. Strikingly, an ultralow field of ~0.001 V/Å along $[\bar{1}10]$ is sufficient to render the D-type phase more stable than its C-type counterpart, enabling an electric-field-induced C-to-D transition. The

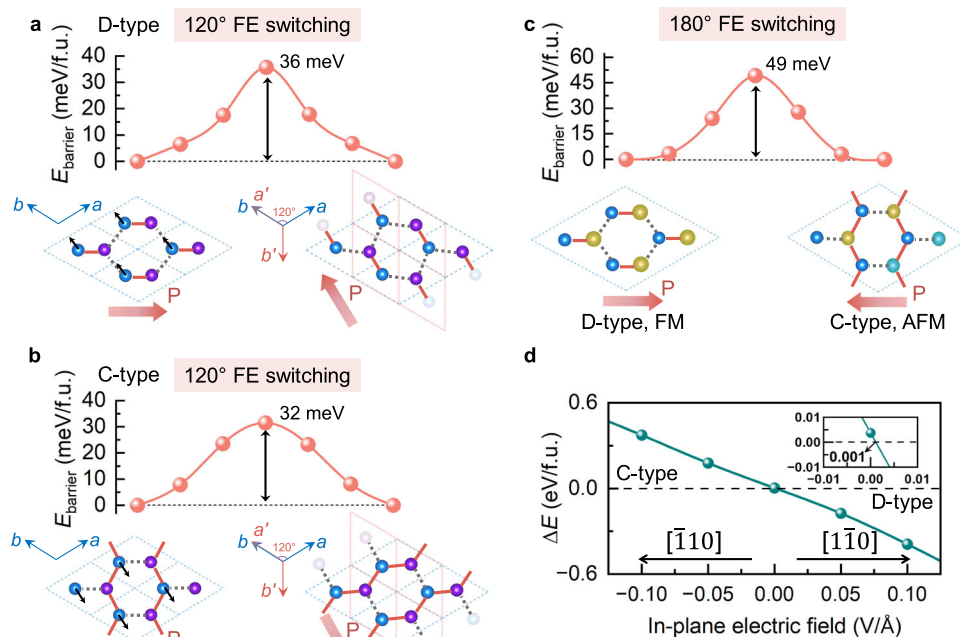


Fig. 4 | Energy barriers for ferroelectric (FE) switching pathways and electric-field-induced magnetic transitions. **a** Energy barrier (E_{barrier} in meV/f.u.) for the 120° ferroelectric switching of the D-type $\text{CuMnP}_2\text{Se}_6$. **b** Energy barrier (in meV/f.u.) for the 120° ferroelectric switching of the C-type $\text{CuMnP}_2\text{Se}_6$. **c** Energy barrier (in meV/f.u.) for the D-to-C phase transition, corresponding to a 180° reversal of polarization. The insets show the spin density distributions of the two phases (with

P and Se atoms omitted), revealing ferromagnetic (FM) and antiferromagnetic (AFM) ground states, respectively. **d** Energy difference between the C-type and D-type phases, defined as $\Delta E = E(\text{D-type}) - E(\text{C-type})$, as a function of the applied in-plane electric field. Gradient orange hollow arrows indicate the direction of in-plane polarization. Positive and negative electric fields correspond to the $[\bar{1}10]$ and $[\bar{1}\bar{1}0]$ crystallographic directions, respectively.

critical field lies well within experimentally accessible limits^{8,65}, highlighting the potential of $\text{CuMnP}_2\text{Se}_6$ for practical multifunctional device applications.

In summary, we identify four stable 2D multiferroics of the form $\text{XMnP}_2(\text{S/Se})_6$ ($X = \text{Cu, Au}$), each exhibiting two inequivalent C_2 -symmetric polar phases with opposite in-plane polarization directions, based on crystal structure prediction and high-throughput first-principles calculations. Jahn-Teller activity and symmetry-forbidden s - d coupling are identified as the two primary driving forces for the emergence of in-plane polarization in these systems. Notably, in $\text{CuMnP}_2\text{Se}_6$, the two polar phases differ not only in their polarization orientation but also in their magnetic ground states (FM and AFM), enabling electric-field-driven switching of both ferroic orders. The ultralow energy barrier (~49 meV/f.u.) and critical switching field ($\sim 0.001 \text{ V } \text{\AA}^{-1}$) render these coupled transitions experimentally accessible. These findings demonstrate a viable route for electric control of magnetism in single-phase 2D multiferroics and provide new insights into ferroic coupling in the BMTP family, with promising implications for low-power, non-volatile device applications. This work also opens a promising avenue for exploring ultralow-field electric control of magnetism enabled by Jahn-Teller-assisted phase-transition mechanisms.

Methods

First-principles calculations were performed based on density functional theory⁶⁶ via the Vienna Ab initio Simulation Package (VASP)^{67,68} using the generalized gradient approximation with the projected augmented wave method⁶⁹ with a 450 eV plane-wave energy cutoff. The Perdew-Burke-Ernzerhof function was adopted for the exchange-correlation functional⁷⁰. A Γ -centered k -point grid of $11 \times 11 \times 1$ was applied for sampling the Brillouin zone. The energy and force threshold for structural optimization are 10^{-7} eV and $0.001 \text{ eV/\AA}$, respectively. An effective Hubbard U of 4 eV for Mn 3d, Cu 3d and Tc 4d orbitals^{71,72} was applied following the Dudarev formulation⁷³. Phonon spectra were

calculated using the finite displacement method implemented in the PHONOPY code⁷⁴. The temperature-dependent effective potential method was used to determine the influence of thermal effects on phonon spectra⁷⁵. Ab initio molecular dynamics simulations were performed utilizing the Nosé-Hoover chain thermostat for 5 ps with a time step of 1 fs with $3 \times 3 \times 1$ supercell. Climbing image nudged elastic band calculations⁷⁶ were performed to determine the energy barriers for ferroelectric switching. The polarization was estimated using the Berry phase approach⁷⁷. An in-plane electric field was applied by imposing a finite homogeneous field using the perturbation expression after discretization approach^{78,79}. The maximally localized Wannier functions are constructed by using the WANNIER90 code⁸⁰.

The particle-swarm optimization method within the evolutionary algorithm, as implemented in the CALYPSO code^{46–51}, was employed to search for the lowest-energy structures of the $\text{CuMnP}_2\text{Se}_6$ monolayer. Unit cells containing one and two formula units were considered. In the initial generation, random structures with specific symmetries were constructed by generating atomic coordinates through crystallographic symmetry operations. Local optimizations were carried out using the VASP code^{67,68} with the conjugate-gradient method and terminated when the Gibbs free-energy change was smaller than $1 \times 10^{-4} \text{ eV}$ per cell. After each generation, 40% of the low-energy structures were selected to construct the next generation through particle swarm optimization, while the remaining 60% were randomly generated. A structure fingerprinting technique based on the bond-characterization matrix was employed to avoid duplication and enhance population diversity, which is crucial for efficient global searching. In total, the search typically sampled 1200 structures (40 generations $\times$ 30 structures).

Data availability

All data generated in this study are provided in the Supplementary Information.

References

- Matsukura, F., Tokura, Y. & Ohno, H. Control of magnetism by electric fields. *Nat. Nanotechnol.* **10**, 209–220 (2015).
- Ramesh, R. & Manapatruni, S. Electric field control of magnetism. *Proc. R. Soc. A* **477**, 20200942 (2021).
- Fert, A., Ramesh, R., Garcia, V., Casanova, F. & Bibes, M. Electrical control of magnetism by electric field and current-induced torques. *Rev. Mod. Phys.* **96**, 015005 (2024).
- Ghidini, M. et al. Non-volatile electrically-driven repeatable magnetization reversal with no applied magnetic field. *Nat. Commun.* **4**, 1453 (2013).
- Cai, K. et al. Electric field control of deterministic current-induced magnetization switching in a hybrid ferromagnetic/ferroelectric structure. *Nat. Mater.* **16**, 712–716 (2017).
- Gong, C., Kim, E. M., Wang, Y., Lee, G. & Zhang, X. Multiferroicity in atomic van der Waals heterostructures. *Nat. Commun.* **10**, 2657 (2019).
- Cherifi, R. et al. Electric-field control of magnetic order above room temperature. *Nat. Mater.* **13**, 345–351 (2014).
- Jiang, S., Shan, J. & Mak, K. F. Electric-field switching of two-dimensional van der Waals magnets. *Nat. Mater.* **17**, 406–410 (2018).
- Huang, B. et al. Electrical control of 2D magnetism in bilayer CrI₃. *Nat. Nanotechnol.* **13**, 544–548 (2018).
- Liu, B., Feng, J., Xiang, H., Dai, Z. & Guo, Z.-X. Efficient electric field control of the magnetic phase in bilayer magnets via interlayer hopping modulation. *Phys. Rev. B* **111**, L140401 (2025).
- Schmid, H. Multi-ferroic magnetoelectrics. *Ferroelectrics* **162**, 317–338 (1994).
- Qi, J., Wang, H., Chen, X. & Qian, X. Two-dimensional multiferroic semiconductors with coexisting ferroelectricity and ferromagnetism. *Appl. Phys. Lett.* **113**, 043102 (2018).
- Huang, C. et al. Prediction of intrinsic ferromagnetic ferroelectricity in a transition-metal halide monolayer. *Phys. Rev. Lett.* **120**, 147601 (2018).
- Zhao, Y. et al. Surface vacancy-induced switchable electric polarization and enhanced ferromagnetism in monolayer metal trihalides. *Nano Lett.* **18**, 2943–2949 (2018).
- Hu, D. et al. Two-dimensional 5d multiferroic W₃Cl₈: Breathing kagome lattice and tunable magneto-optical Kerr effect. *Phys. Rev. B* **109**, 014433 (2024).
- Tan, H. et al. Two-dimensional ferromagnetic-ferroelectric multiferroics in violation of the d⁰ rule. *Phys. Rev. B* **99**, 195434 (2019).
- Ding, N., Chen, J., Dong, S. & Stroppa, A. Ferroelectricity and ferromagnetism in a VOI₂ monolayer: Role of the Dzyaloshinskii-Moriya interaction. *Phys. Rev. B* **102**, 165129 (2020).
- Hao, K.-R. et al. Ferroelectric and room-temperature ferromagnetic semiconductors in the 2D M₁M₂Ge₂X₆ family: first-principles and machine learning investigations. *J. Phys. Chem. Lett.* **12**, 10040–10051 (2021).
- Yang, Y. et al. Two-dimensional organic–inorganic room-temperature multiferroics. *J. Am. Chem. Soc.* **144**, 14907–14914 (2022).
- Jiang, J. et al. Ferroelectricity-driven magnetism in a metal halide monolayer. *Phys. Rev. Lett.* **134**, 196801 (2025).
- Wang, Z. & Dong, S. Alterferroicity with seesaw-type magnetoelectricity. *Proc. Natl. Acad. Sci.* **120**, e2305197120 (2023).
- Tang, C., Zhang, L., Sanvito, S. & Du, A. Enabling room-temperature triferroic coupling in dual transition-metal dichalcogenide monolayers via electronic asymmetry. *J. Am. Chem. Soc.* **145**, 2485–2491 (2023).
- Huang, C. et al. Toward room-temperature electrical control of magnetic order in multiferroic van der Waals materials. *Nano Lett.* **22**, 5191–5197 (2022).
- Liu, X., Pyatakova, A. P. & Ren, W. Magnetoelectric coupling in multiferroic bilayer VS₂. *Phys. Rev. Lett.* **125**, 247601 (2020).
- Yu, G. et al. Ferroelectrically switchable magnetic multistates in MnBi₂Te₄ (Bi₂Te₃)_n and MnSb₂Te₄ (Sb₂Te₃)_n (n = 0, 1) thin films. *Phys. Rev. B* **108**, 014106 (2023).
- Su, Y. et al. Van der Waals multiferroic tunnel junctions. *Nano Lett.* **21**, 175–181 (2020).
- Xu, X. & Yang, L. Enhanced magnetoelectric coupling in two-dimensional hybrid multiferroic heterostructures. *Phys. Rev. B* **110**, 165408 (2024).
- Pizzocchero, F. et al. The hot pick-up technique for batch assembly of van der Waals heterostructures. *Nat. Commun.* **7**, 11894 (2016).
- Xu, M. et al. Electrical control of magnetic phase transition in a type-I multiferroic double-metal trihalide monolayer. *Phys. Rev. Lett.* **124**, 067602 (2020).
- Qu, Z. et al. Strong magnetoelectric coupling in a double transition metal dichalcogenide monolayer. *Phys. Rev. B* **110**, L161408 (2024).
- Chen, J. et al. Control of magnetic transitions via interlayer engineering in ferroelectric H₂O–OH systems. *Nat. Commun.* **16**, 4809 (2025).
- Susner, M. A., Chyashnavichyus, M., McGuire, M. A., Ganesh, P. & Maksymovych, P. Metal thio- and selenophosphates as multifunctional van der Waals layered materials. *Adv. Mater.* **29**, 1602852 (2017).
- Si, M. et al. Room-temperature electrocaloric effect in layered ferroelectric CuInP₂S₆ for solid-state refrigeration. *ACS Nano* **13**, 8760–8765 (2019).
- Brehm, J. A. et al. Tunable quadruple-well ferroelectric van der Waals crystals. *Nat. Mater.* **19**, 43–48 (2020).
- Zhou, S. et al. Van der Waals layered ferroelectric CuInP₂S₆: Physical properties and device applications. *Front. Phys.* **16**, 13301 (2021).
- Ma, X.-Y. et al. Large family of two-dimensional ferroelectric metals discovered via machine learning. *Sci. Bull.* **66**, 233–242 (2021).
- Duan, X., Wang, H., Chen, X. & Qi, J. Multiple polarization phases and strong magnetoelectric coupling in the layered transition metal phosphorus chalcogenides TMP₂X₆ (T = Cu, Ag; M = Cr, V; X = S, Se) by controlling the interlayer interaction and dimension. *Phys. Rev. B* **106**, 115403 (2022).
- Zhang, D., Li, A., Chen, X., Zhou, W. & Ouyang, F. Tuning valley splitting and magnetic anisotropy of multiferroic CuMP₂X₆ (M = Cr, V; X = S, Se) monolayer. *Phys. Rev. B* **105**, 085408 (2022).
- Yadav, A., Stojić, N. & Binggeli, N. Inducing out-of-plane magnetic anisotropy by chemical substitutions in the multiferroic CuCrP₂S₆ monolayer. *Phys. Rev. B* **110**, 144404 (2024).
- Liu, F. et al. Room-temperature ferroelectricity in CuInP₂S₆ ultrathin flakes. *Nat. Commun.* **7**, 12357 (2016).
- Io, W. F. et al. Direct observation of intrinsic room-temperature ferroelectricity in 2D layered CuCrP₂S₆. *Nat. Commun.* **14**, 7304 (2023).
- Yang, Y. et al. 2D ferroelectric AgInP₂Se₆ for ultra-steep slope transistor with SS below 10 mV decade⁻¹. *Adv. Electron. Mater.* **2400685** (2025).
- Lai, Y. et al. Two-dimensional ferromagnetism and driven ferroelectricity in van der Waals CuCrP₂S₆. *Nanoscale* **11**, 5163–5170 (2019).
- Zhang, X. et al. Origin of versatile polarization state in CuInP₂S₆. *Phys. Rev. B* **108**, L161406 (2023).
- Wang, Q. et al. Magnetic and multiferroic properties of two-dimensional FePX₃ and CuFeP₂X₆ (X = S, Se, and Te). *ACS Appl. Electron. Mater.* **7**, 660–666 (2025).
- Wang, Y., Lv, J., Zhu, L. & Ma, Y. Crystal structure prediction via particle-swarm optimization. *Phys. Rev. B* **82**, 094116 (2010).
- Wang, Y., Lv, J., Zhu, L. & Ma, Y. Calypso: A method for crystal structure prediction. *Comput. Phys. Commun.* **183**, 2063–2070 (2012).

48. Wang, Y. et al. An effective structure prediction method for layered materials based on 2D particle swarm optimization algorithm. *J. Chem. Phys.* **137**, 224108 (2012).
49. Wang, Y. & Ma, Y. Perspective: Crystal structure prediction at high pressures. *J. Chem. Phys.* **140**, 040901 (2014).
50. Gao, B. et al. Interface structure prediction via CALYPSO method. *Sci. Bull.* **64**, 301–309 (2019).
51. Shao, X. et al. A symmetry-orientated divide-and-conquer method for crystal structure prediction. *J. Chem. Phys.* **156**, 014105 (2022).
52. Bhattarai, R., Minch, P., Liang, Y., Zhang, S. & Rhone, T. D. Strain-induced topological phase transition in ferromagnetic janus monolayer MnSbBiS₂Te₂. *Phys. Chem. Chem. Phys.* **26**, 10111–10119 (2024).
53. Sun, Y., Zhuo, Z., Wu, X. & Yang, J. Room-temperature ferromagnetism in two-dimensional Fe₂Si nanosheet with enhanced spin-polarization ratio. *Nano Lett.* **17**, 2771–2777 (2017).
54. Sivadas, N., Daniels, M. W., Swendsen, R. H., Okamoto, S. & Xiao, D. Magnetic ground state of semiconducting transition-metal trichalcogenide monolayers. *Phys. Rev. B* **91**, 235425 (2015).
55. Bennett, D., Jankowski, W. J., Chaudhary, G., Kaxiras, E. & Slager, R.-J. Theory of polarization textures in crystal supercells. *Phys. Rev. Res.* **5**, 033216 (2023).
56. He, R. et al. Unconventional ferroelectric domain switching dynamics in CuInP₂S₆ from first principles. *Phys. Rev. B* **108**, 024305 (2023).
57. Xu, B. et al. Monolayer AgBiP₂Se₆: an atomically thin ferroelectric semiconductor with out-plane polarization. *Nanoscale* **9**, 8427–8434 (2017).
58. Seleznev, D., Singh, S., Bonini, J., Rabe, K. M. & Vanderbilt, D. Cyclic ferroelectric switching and quantized charge transport in CuInP₂S₆. *Phys. Rev. B* **108**, L180101 (2023).
59. Wu, M. & Zeng, X. C. Intrinsic ferroelasticity and/or multiferroicity in two-dimensional phosphorene and phosphorene analogues. *Nano Lett.* **16**, 3236–3241 (2016).
60. Wu, M. & Zeng, X. C. Bismuth oxychalcogenides: a new class of ferroelectric/ferroelastic materials with ultra high mobility. *Nano Lett.* **17**, 6309–6314 (2017).
61. Liu, C. et al. First-principles study of ferroelectricity, anti-ferroelectricity, and ferroelasticity in two-dimensional γ-ALOOH. *Phys. Rev. B* **107**, L121402 (2023).
62. Duan, C.-G., Jaswal, S. S. & Tsymbal, E. Y. Predicted magnetoelectric effect in Fe/BaTiO₃ multilayers: Ferroelectric control of magnetism. *Phys. Rev. Lett.* **97**, 047201 (2006).
63. Xu, B., Garcia, V., Fusil, S., Bibes, M. & Bellaiche, L. Intrinsic polarization switching mechanisms in BiFeO₃. *Phys. Rev. B* **95**, 104104 (2017).
64. Wang, Y. et al. A stable rhombohedral phase in ferroelectric Hf(Zr)_{1-x}O₂ capacitor with ultralow coercive field. *Science* **381**, 558–563 (2023).
65. Kang, K. et al. Switchable moiré potentials in ferroelectric WTe₂/WSe₂ superlattices. *Nature Nanotechnology* **18**, 861–866 (2023).
66. Kohn, W. & Sham, L. J. Self-consistent equations including exchange and correlation effects. *Phys. Rev.* **140**, A1133 (1965).
67. Kresse, G. & Hafner, J. Ab initio molecular dynamics for open-shell transition metals. *Phys. Rev. B* **48**, 13115 (1993).
68. Kresse, G. & Furthmüller, J. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **54**, 11169 (1996).
69. Blöchl, P. E. Projector augmented-wave method. *Phys. Rev. B* **50**, 17953 (1994).
70. Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **77**, 3865 (1996).
71. Swamynadhan, M. & Ghosh, S. Designing multifunctional two-dimensional layered transition metal phosphorous chalcogenides. *Phys. Rev. Mater.* **5**, 054409 (2021).
72. Tesch, R. & Kowalski, P. M. Hubbard U parameters for transition metals from first principles. *Phys. Rev. B* **105**, 195153 (2022).
73. Dudarev, S. L., Botton, G. A., Savrasov, S. Y., Humphreys, C. & Sutton, A. P. Electron-energy-loss spectra and the structural stability of nickel oxide: An LSDA+ U study. *Phys. Rev. B* **57**, 1505 (1998).
74. Togo, A., Oba, F. & Tanaka, I. First-principles calculations of the ferroelastic transition between rutile-type and CaCl₂-type SiO₂ at high pressures. *Phys. Rev. B* **78**, 134106 (2008).
75. Knoop, F. et al. TDEP: Temperature dependent effective potentials. *J. Open Source Softw.* **9**, 6150 (2024).
76. Henkelman, G., Uberuaga, B. P. & Jónsson, H. A climbing image nudged elastic band method for finding saddle points and minimum energy paths. *J. Chem. Phys.* **113**, 9901–9904 (2000).
77. King-Smith, R. D. & Vanderbilt, D. Theory of polarization of crystalline solids. *Phys. Rev. B* **47**, 1651–1654 (1993).
78. Ding, W. et al. Prediction of intrinsic two-dimensional ferroelectrics in In₂Se₃ and other III₂-VI₃ van der Waals materials. *Nat. Commun.* **8**, 14956 (2017).
79. Nunes, R. & Gonze, X. Berry-phase treatment of the homogeneous electric field perturbation in insulators. *Phys. Rev. B* **63**, 155107 (2001).
80. Mostofi, A. A. et al. An updated version of wannier90: A tool for obtaining maximally-localised wannier functions. *Comput. Phys. Commun.* **185**, 2309–2310 (2014).

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M.X., X.Z., Y.L., and Y.W. conceived and designed the project and directed the calculations. J.C. and M.X. performed structure prediction and first-principles calculations. M.X., X.Z., J.F., and Y.J. analyzed the data. M.X. and Y.W. wrote the manuscript with input from all other authors. This manuscript reflects the contributions of all authors.

Competing interests

The authors declare no competing interests.

Additional information

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